

Data Flow Diagram

Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Team ID	SWTID-2026-1254
Team Lead	Kamalesh S
Team Members	Tamilselvan P, Tamilselvam K, Mugunthan S, Suriya L

Input Flow

User enters story prompt, character name, setting, tone, and art style through index.html.

Backend Flow

The POST request reaches the FastAPI backend in routes.py. Form input is processed using FastAPI Form parameters; JSON requests are validated using the PromptRequest Pydantic schema.

AI Flow

routes.py calls generate_outline() in gemini_flash.py → generate_story() in gemini_pro.py → generate_image() in image_generator.py for each panel.

Layout Flow

build_comic_layout() in layout_builder.py matches each generated image with its corresponding panel story and creates the final layout structure.

Export Flow

save_pdf() in exporters.py uses FPDF to place panel images and narrative text into a multi-page PDF and saves it under static/exports.

Presentation Flow

Generated data is passed to comic_preview.html; successful export redirects the user to export_success.html.